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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 6	
TITLE : Inner and Anonymous Classes	

Problem Statement

Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives

To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Programme

1.Create a Vehicle program where an inner class displays vehicle details and anonymous class performs an action.

Code:

```
interface Action {
    void performAction();
}
class Vehicle {
    String name;
    String model;

    Vehicle(String name, String model) {
        this.name = name;
        this.model = model;
    }

    class VehicleDetails {
        void display() {
            System.out.println("Vehicle Name : " + name);
            System.out.println("Vehicle Model: " + model);
        }
    }
}
public class VehicleDemo {
```

```
    public static void main(String[] args) {

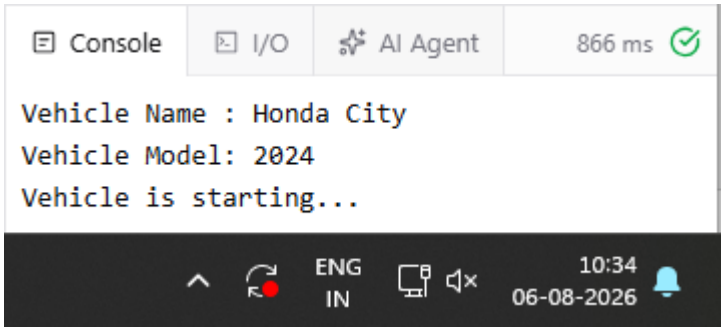
        Vehicle v = new Vehicle("Honda City", "2024");

        Vehicle.VehicleDetails details = v.new VehicleDetails();
        details.display();

        Action action = new Action() {
            public void performAction() {
                System.out.println("Vehicle is starting...");
            }
        };

        action.performAction();
    }
}
```

Output



2. Develop a Food Delivery Application where an inner class handles order details and anonymous classes handle delivery status updates.

Code:

```
interface DeliveryStatus {
    void status();
}
class FoodOrder {
    String customerName;
    String foodItem;

    FoodOrder(String customerName, String foodItem) {
        this.customerName = customerName;
        this.foodItem = foodItem;
    }
    class OrderDetails {
        void displayOrder() {
            System.out.println("Customer Name : " + customerName);
```

```
            System.out.println("Food Item     : " + foodItem);
        }
    }
}

public class FoodDeliveryApp {
    public static void main(String[] args) {

        FoodOrder order = new FoodOrder("Saswati", "Chicken Burger");

        FoodOrder.OrderDetails details = order.new OrderDetails();
        details.displayOrder();

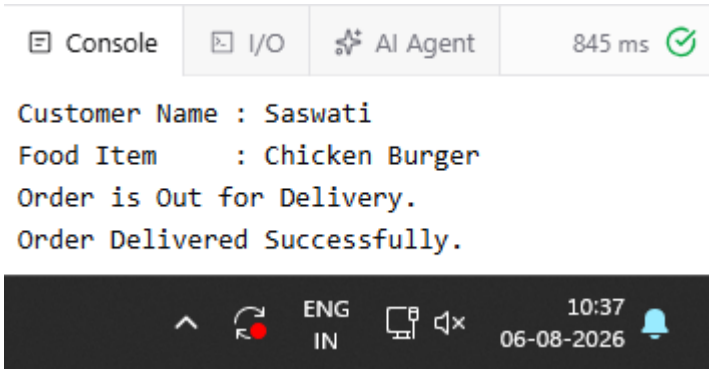
        DeliveryStatus delivery = new DeliveryStatus() {
            public void status() {
                System.out.println("Order is Out for Delivery.");
            }
        };

        delivery.status();

        DeliveryStatus delivered = new DeliveryStatus() {
            public void status() {
                System.out.println("Order Delivered Successfully.");
            }
        };

        delivered.status();
    }
}
```

Output:



Conclusion

This experiment helped in understanding the use of inner classes and anonymous classes in Java. Inner classes improve encapsulation by keeping related classes together and allowing access to the outer class members. Anonymous classes reduce code by providing a one-time implementation without creating a separate class. These concepts are useful in Java programming, especially for GUI applications and event handling, making the code more organized and efficient.